

Dualizing modules and functors

SGA 2, Exposé IV

English draft

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About this draft

This is an unofficial English translation of Exposé IV, “Dualizing modules and functors”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ IV

Dualizing modules and functors

1. Generalities on functors of modules

Let A be a commutative noetherian ring,
 $\underline{\mathcal{C}}$ the category of A -modules of finite type,
 $\underline{\mathcal{C}}'$ the category of arbitrary A -modules,
 $\underline{\mathcal{A}b}$ the category of abelian groups.

The aim of this paragraph is the study of certain properties of functors $T : \underline{\mathcal{C}}^\circ \rightarrow \underline{\mathcal{A}b}$ (assumed additive).

Let us remark that if $M \in \text{Ob } \underline{\mathcal{C}}$, $T(M)$ can be equipped in a canonical way with an A -module structure which is the following: if f_M is the homothety of M associated to $f \in A$, A operates on $T(M)$ by $f_{T(M)}$. In other words, T factors as:

$$\begin{array}{ccc} \underline{\mathcal{C}}^\circ & \xrightarrow{T} & \underline{\mathcal{A}b} \\ & \searrow T_\circ \quad \nearrow & \\ & \underline{\mathcal{C}}' & \end{array}$$

where $\underline{\mathcal{C}}' \rightarrow \underline{\mathcal{A}b}$ is the canonical functor.

In what follows, $T(M)$ will be considered as equipped with its A -module structure.

Composing with the isomorphism $M \xrightarrow{\sim} \text{Hom}_A(A, M)$ the morphism $\text{Hom}_A(A, M) \rightarrow \text{Hom}_A(T(M), T(A))$, one obtains the following morphisms which are deduced from one another in an evident manner:

$$\begin{aligned} M &\longrightarrow \text{Hom}_A(T(M), T(A)), \\ M \times T(M) &\longrightarrow T(A), \\ T(M) &\longrightarrow \text{Hom}_A(M, T(A)), \end{aligned}$$

which defines a morphism φ_T of contravariant functors:

$$\varphi_T : T \longrightarrow \text{Hom}_A(M, T(A)).$$

PROPOSITION 1.1. *The following two properties are equivalent:*

- (i) φ_T is an isomorphism of functors.
- (ii) T is left exact.

The implication (i) $\Rightarrow$ (ii) is trivial.

The implication (ii) $\Rightarrow$ (i) follows from the fact that for a morphism $u : F \rightarrow F'$ of two additive left exact functors F and F' from $\underline{\mathcal{C}}^\circ$ to $\underline{\mathcal{A}b}$, if $u(A)$ is an isomorphism, u is an isomorphism (one uses the fact that A is noetherian and hence that every A -module of finite type is of finite presentation).

REMARK 1.2. This shows in particular that the functors $T : \underline{\mathcal{C}}^\circ \rightarrow \underline{\mathcal{A}b}$ which are representable are the functors which commute with arbitrary projective limits (over a preordered set, not necessarily filtered).

If $\underline{\text{Hom}}(\underline{\mathcal{C}}^\circ, \underline{\mathcal{A}b})_g$ denotes the full subcategory of $\underline{\text{Hom}}(\underline{\mathcal{C}}^\circ, \underline{\mathcal{A}b})$ whose objects are the left exact functors, one has proved the equivalence of categories

$$\underline{\mathcal{C}}' \xrightarrow{\sim} \underline{\text{Hom}}(\underline{\mathcal{C}}^\circ, \underline{\mathcal{A}b})_g$$

via the functors quasi-inverse to one another

$$H \longmapsto \text{Hom}_A(\ , H)$$

and

$$T(A) \longleftarrow T.$$

Let now J be an ideal of A , $Y = V(J) \subset \text{Spec } A$, and let us denote by $\underline{C}_Y$ the full subcategory of $\underline{C}$ whose objects are the A -modules of finite type M such that $\text{Supp } M \subset Y$. One has:

$$\underline{C}_Y = \bigcup_n \underline{C}^{(n)},$$

where $\underline{C}^{(n)}$ is the full subcategory of $\underline{C}_Y$ of those modules M such that $J^n M = 0$.

PROPOSITION 1.3. *With the same notations as above, let $T : \underline{C}_Y \rightarrow \underline{\text{Ab}}$ be a functor. To $H = \varinjlim T(A/J^n)$ ¹ is associated a natural morphism*

$$\varphi_T : T \longrightarrow \text{Hom}_A(\ , H),$$

and the following conditions are equivalent:

- (i) φ_T is an isomorphism.
- (ii) The functor T is left exact.

PROOF. a) Definition of φ_T .

Let $M \in \text{Ob } \underline{C}_Y$. There exists an integer n such that $J^n M = 0$. Then M is an A/J^n -module, and if T_n denotes the restriction of T to $\underline{C}^{(n)}$, one knows how to define the morphism

$$T_n \longrightarrow \text{Hom}_A(\ , H_n), \text{ where } H_n = T(A/J^n);$$

$$\text{whence } T(M) = T_n(M) \longrightarrow \text{Hom}_A(M, \varinjlim H_n) = \text{Hom}_A(M, H)$$

$$\text{and } \varphi_T : T \longrightarrow \text{Hom}_A(\ , H).$$

b) Equivalence of (i) and (ii).

It is clear that (i) implies (ii). Suppose (ii) holds and let $M \in \text{Ob } \underline{C}^{(n)}$. One has seen that $T_n(M) \xrightarrow{\sim} \text{Hom}_A(M, H_n)$, hence for every integer $n' > n$ one has

$$T(M) = T_n(M) = T_{n'}(M) = \varinjlim T_n(M)$$

$$\text{and } T(M) = \varinjlim \text{Hom}_A(M, H_n).$$

As one is dealing here with filtered inductive limits, one also has the isomorphism

$$\varinjlim \text{Hom}_A(M, H_n) \xrightarrow{\sim} \text{Hom}_A(M, \varinjlim H_n) = \text{Hom}_A(M, H).$$

If $\underline{C}'_Y$ denotes the category of A -modules, with support contained in Y , but not necessarily of finite type, one still has the natural equivalence of categories: $\underline{C}'_Y \xrightarrow{\sim} \text{Hom}(\underline{C}_Y^\circ, \underline{\text{Ab}})_g$. $\square$

Application. The notations being the same as above, let

$$T^* : \underline{C}_Y^\circ \longrightarrow \underline{\text{Ab}}$$

be an exact ∂ -functor. For every $i \in \mathbf{Z}$, one sets $H_n^i = T^i(A/J^n)$ and $H^i = \varinjlim H_n^i$.

THEOREM 1.4. *Let $n \in \mathbf{Z}$. If there exists $i_0 \in \mathbf{Z}$ such that $T^i = 0$ for every $i < i_0$, then the following three conditions are equivalent:*

- (i) $T^i = 0$ for every $i < n$.
- (ii) $H^i = 0$ for every $i < n$.
- (iii) There exists a module M_0 of $\underline{C}_Y$ such that $\text{Supp } M_0 = Y$ and $T^i(M_0) = 0$ for every $i < n$.

¹**N.D.E.** the definition of H is implicit in the original text.

PROOF. It is evident that (i) implies (ii) and (iii) (one takes $M_0 = A/J$). Let us show by induction on n that (ii) implies (i); this is true for $n = i_0$, and one assumes it proved up to n . Suppose then that $H^i = 0$ for every $i < n + 1$; by the induction hypothesis one has $T^i = 0$ for $i < n$, but $T^{n-1} = 0$ implies that T^n is a left exact functor and

$$T^n \xrightarrow{\sim} \text{Hom}_A(_, H^n) = 0.$$

Let us then show that (iii) implies (ii). This is still true for $n = i_0$; suppose it proved up to n : let M_0 be an A -module of $\underline{C}_Y$ such that $\text{Supp } M_0 = Y$ and $T^i(M_0) = 0$ for every $i < n + 1$; by the induction hypothesis one then has $H^i = 0$ for every $i < n$; it remains to show that $H^n = 0$. But “ $H^i = 0$ for every $i < n$ ” implies that $T^{n-1} = 0$ and hence that $T^n \xrightarrow{\sim} \text{Hom}_A(_, H^n)$. One then has

$$\text{Ass } H^n = \text{Ass Hom}(M_0, H^n) = \text{Supp } M_0 \cap \text{Ass } H^n = \text{Ass } H^n$$

since

$$\text{Ass } H^n \subset \text{Supp } H^n \subset Y = \text{Supp } M_0.$$

Whence $T^n(M_0) = 0 \Leftrightarrow \text{Ass } H^n = \emptyset \Leftrightarrow H^n = 0$; which completes the proof. $\square$

2. Characterization of exact functors

The ring A is always assumed noetherian and commutative. The notations are those of proposition 1.3:

$$Y = V(J), \quad T : \underline{C}_Y^\circ \longrightarrow \underline{\text{Ab}}, \quad H = \varinjlim T(A/J^n),$$

where one assumes that T is a left exact functor, whence:

$$T(M) \xrightarrow{\sim} \text{Hom}_A(M, H).$$

PROPOSITION 2.1. *The following properties are equivalent:*

- (i) *The functor T is exact,*
- (ii) *H is injective in $\underline{C}'$.*

PROOF. It obviously suffices to show that (i) implies (ii), that is to say to prove that if the restriction to $\underline{C}_Y$ of the functor $\text{Hom}_A(_, H)$ is an exact functor, H is injective. But as A is noetherian, to show that H is injective, one may restrict oneself to proving that every homomorphism $f : N \rightarrow H$ with source an A -module N of finite type, a submodule of an A -module M of finite type, extends to a homomorphism $\bar{f} : M \rightarrow H$.

The definition of H and the fact that N is of finite type imply that there exists an integer n such that $J^n \cdot f(N) = 0$. Let us then equip M and N with the J -adic topology. The J -adic topology of N is equivalent to the topology induced by the J -adic topology of M (Krull's theorem). There thus exists $V = J^k \cdot M$ such that

$$U = V \cap N \subset J^n N.$$

One then has the factorization

$$\begin{array}{ccc} N & \longrightarrow & N/U, \\ f \downarrow & \swarrow u & \\ & & H \end{array}$$

N/U and M/V are in $\underline{C}_Y$. The hypothesis thus allows one to extend u to $\bar{u}$

$$\begin{array}{ccc} N/U & \hookrightarrow & M/V, \\ u \downarrow & \swarrow \bar{u} & \\ & & H \end{array}$$

and $M \rightarrow M/V \xrightarrow{\bar{u}} H$ gives the desired extension $\bar{f}$. $\square$

COROLLARY 2.2. *Let K be an injective A -module; then the submodule $H_J^0(K)$ of K consisting of the elements annihilated by a suitable power of J is injective.*

PROOF. It suffices to check that the restriction to $\underline{C}_Y$ of the functor $\text{Hom}_A(, H_J^0(K))$ is an exact functor. Now let $M \in \text{Ob } \underline{C}_Y$; there exists k such that $J^k \cdot M = 0$, and the inclusion

$$\text{Hom}_A(, H_J^0(K)) \longrightarrow \text{Hom}_A(M, K)$$

is then an isomorphism. Hence the result, since $\text{Hom}_A(, K)$ is exact. $\square$

3. Study of the case where T is left exact and $T(M)$ is of finite type for every M

Let as above

$$T : \underline{C}_Y^\circ \longrightarrow \underline{\text{Ab}},$$

one assumes henceforth that T is left exact and that one has the factorization

$$\begin{array}{ccc} \underline{C}_Y^\circ & \xrightarrow{T} & \underline{\text{Ab}} \\ & \searrow & \nearrow \\ & \underline{C}_Y & \end{array}$$

where as above, $\underline{C}_Y \rightarrow \underline{\text{Ab}}$ is the forgetful functor. One thus knows how to define $T(T(M)) = T \circ T(M)$, and the canonical morphism

$$M \longrightarrow \text{Hom}_A(\text{Hom}_A(M, H), H)$$

defines a morphism

$$M \longrightarrow T \circ T(M).$$

PROPOSITION 3.1. *The ring A still being assumed noetherian, if one makes the additional hypothesis that A/J is artinian, the following conditions are equivalent:*

- (i) *T is left exact and, for every $M \in \text{Ob } \underline{C}_Y$, $T(M)$ is of finite type and $M \rightarrow T \circ T(M)$ is an isomorphism.*
- (ii) *T is exact and, for every residue field k associated to a maximal ideal containing J , one has $T(k) \xrightarrow{\sim} k$.*
- (iii) *One has $T \xrightarrow{\sim} \text{Hom}_A(, H)$ with H injective and, for every k as in (ii), one has $\text{Hom}_A(k, H) \xrightarrow{\sim} k$.*
- (iv) *T is exact and, for every $M \in \text{Ob } \underline{C}_Y$, one has $\text{long } T(M) = \text{long } M$.*

PROOF. One has already shown the equivalence of (ii) and (iii) (prop. 2.1). Let us show that (ii) implies (iv): first if $M \in \text{Ob } \underline{C}_Y$, as M is an A/J^n -module with A/J^n artinian, $\text{long } M$ is finite. Let us argue by induction on the length of M . Condition (iv) is true if $\text{long } M = 1$, because then M is a residue field to which (ii) applies. If $\text{long } M > 1$, there exists a submodule M' of M such that $M' \neq 0$ and $\text{long } M' < \text{long } M$. Let us then form the exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

As T is exact, one has the sequence

$$0 \longrightarrow T(M') \longrightarrow T(M) \longrightarrow T(M'') \longrightarrow 0,$$

and $\text{long } T(M) = \text{long } T(M') + \text{long } T(M'') = \text{long } M' + \text{long } M'' = \text{long } M$.

(ii) implies (i): As (ii) implies (iv), let M be an A -module of $\underline{C}_Y$; one has $\text{long } T(M) = \text{long } M$; hence $T(M)$ is of finite length and therefore of finite type.

It remains to show that $M \rightarrow T \circ T(M)$ is an isomorphism; one argues again by induction on $\text{long } M$. For $M = k$ this is true. In the general case one writes the commutative diagram whose two rows are exact:

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & T \circ T(M') & \longrightarrow & T \circ T(M) & \longrightarrow & T \circ T(M'') \longrightarrow 0, \end{array}$$

where M' is a submodule of M such that $M' \neq 0$ and $\text{long } M' < \text{long } M$. By the induction hypothesis, the extreme arrows are isomorphisms, hence

$$M \longrightarrow T \circ T(M)$$

is an isomorphism.

(i) implies (ii): let

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be an exact sequence of A -modules of $\underline{\mathbb{C}}_Y$, and let Q be the cokernel of $T(M) \rightarrow T(M')$. Let us apply T to the exact sequence

$$0 \longrightarrow T(M') \longrightarrow T(M) \longrightarrow T(M'') \longrightarrow Q \longrightarrow 0,$$

one obtains:

$$\begin{array}{ccccccc} 0 & \longrightarrow & T(Q) & \longrightarrow & T \circ T(M') & \longrightarrow & T \circ T(M) \\ & & & & \uparrow s & & \uparrow s \\ & & & & M' & \longrightarrow & M \end{array}$$

hence $T(Q) = 0$ and $Q \xrightarrow{\sim} T(T(Q)) = 0$.

On the other hand let k be a residue field, $k = A/\mathfrak{m}$, $J \subset \mathfrak{m}$. One must show that $T(k) \xrightarrow{\sim} k$. For this it suffices to remark that $T(k)$ is a k -vector space. One deduces from this:

$$\begin{aligned} T(k) &\simeq k \oplus V, \\ T(T(k)) &\simeq T(k) \oplus T(V) \simeq k \oplus V \oplus T(V) \simeq k, \end{aligned}$$

whence $V = 0$.

Let us finally show that (iv) implies (iii): it suffices to show that $T(k) \xrightarrow{\sim} k$; but $\text{long } T(k) = \text{long } k = 1$, hence $T(k) = k'$ is a residue field and $\text{Supp } k' = \text{Supp } \text{Hom}_A(k, H) \subset \text{Supp } k$. Hence $k \simeq'$. $\square$

REMARK 3.2. One can show that condition (iv) is equivalent to the condition

(iv)' For every $M \in \text{Ob } \underline{\mathbb{C}}_Y$, one has $\text{long } T(M) = \text{long } M$.

4. Dualizing module. Dualizing functor

DEFINITION 4.1. Let A be a noetherian local ring and $\mathfrak{m}$ its maximal ideal. One calls dualizing functor for A any functor

$$T : \underline{\mathbb{C}}_{\mathfrak{m}}^{\circ} \longrightarrow \underline{\text{Ab}},$$

where one writes $\underline{\mathbb{C}}_{\mathfrak{m}}$ instead of $\underline{\mathbb{C}}_Y$ for $Y = V(\mathfrak{m})$, which satisfies the equivalent conditions of proposition 3.1. One says that an A -module I is dualizing for A if the functor $M \rightarrow \text{Hom}_A(M, I)$ is dualizing.

One can generalize definition 4.1 to the case where one no longer assumes that A is a local ring.

DEFINITION 4.2. Let A be a noetherian ring and let $\overline{\underline{\mathbb{C}}}$ be the full subcategory of $\underline{\mathbb{C}}$ formed of the A -modules of finite length; one calls dualizing functor any functor T , A -linear, from $\overline{\underline{\mathbb{C}}}^{\circ}$ to $\overline{\underline{\mathbb{C}}}$, which is exact and such that the morphism of functors

$$\text{id} \longrightarrow T \circ T$$

is an isomorphism.

One will prove an existence theorem and also that the module I which represents such a functor is locally artinian. One will also show that, for every maximal ideal $\mathfrak{m}$ of A , the $\mathfrak{m}$ -primary component of the socle of I is of length 1.

PROPOSITION 4.3. *Let A and B be two noetherian local rings with maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$, such that B is a finite A -algebra. Then, if I is a dualizing module for A , $\text{Hom}_A(B, I)$ is a dualizing module for B .*

PROOF. Let

$$R : \underline{C}_{\mathfrak{m}_B} \longrightarrow \underline{C}_{\mathfrak{m}_A}$$

be the restriction of scalars functor; it is exact. Let T be a dualizing functor for A ,

$$T : \underline{C}_{\mathfrak{m}_A} \longrightarrow \underline{\text{Ab}};$$

it is exact and, for every $M \in \text{Ob } \underline{C}_{\mathfrak{m}_A}$, the natural morphism $M \rightarrow T \circ T(M)$ is an isomorphism; hence $T \circ R$ is a dualizing functor for B . If I represents T , from the classical formula $\text{Hom}_A(M, I) = \text{Hom}_B(M, \text{Hom}_A(B, I))$, valid for every B -module M , one deduces that $\text{Hom}_A(B, I)$ is a dualizing module for B . $\square$

COROLLARY 4.4. *Let A be a noetherian local ring and $\mathfrak{a}$ an ideal of A ; let $B = A/\mathfrak{a}$. If I is a dualizing module for A , the annihilator of $\mathfrak{a}$ in I is a dualizing module for B .*

LEMMA 4.5. *Let A be a noetherian local ring and I a locally artinian A -module. There exists a canonical isomorphism*

$$I \longrightarrow \hat{I} = I \otimes_A \hat{A}.$$

PROOF. Let I_n be the annihilator of $\mathfrak{m}^n$ in I , where $\mathfrak{m}$ is the maximal ideal of A . To say that I is locally artinian is to say that I is the inductive limit of the I_n and that the latter are of finite length. Now the tensor product commutes with inductive limits, so one is reduced to the case where I is artinian. In this case I is annihilated by a power of the maximal ideal, say $\mathfrak{m}^k$; hence for $p \geq k$, $I \xrightarrow{\sim} I \otimes_A A/\mathfrak{m}^p$, hence $I \xrightarrow{\sim} I \otimes_A \hat{A}$ because A is noetherian and I is of finite type.

One concludes that the restriction of scalars functor from $\hat{A}$ to A and the extension of scalars functor induce *equivalences* quasi-inverse to one another of the category of $\hat{A}$ -locally artinian modules and of the category of A -locally artinian modules. $\square$

PROPOSITION 4.6. *Let A be a noetherian local ring, $\hat{A}$ its completion, I a dualizing module for A (resp. for $\hat{A}$) and J the completion² of I (resp. the A -module obtained by restriction of scalars). Then J is a dualizing module for $\hat{A}$ (resp. for A). Moreover the abelian groups underlying I and J are isomorphic.*

PROOF. One simply remarks that the equivalence between the category of locally artinian A -modules and the category of locally artinian $\hat{A}$ -modules induces an isomorphism between the bifunctors $\text{Hom}_A(,)$ and $\text{Hom}_{\hat{A}}(,)$, and that the characterization of a dualizing functor or module involves only these bifunctors. $\square$

THEOREM 4.7. *Let A be a noetherian local ring.*

- a) *There always exists a dualizing module I .*
- b) *Two dualizing modules are isomorphic (by a non-canonical isomorphism).*
- c) *For a module I to be dualizing, it is necessary and sufficient that it be an injective envelope of the residue field k of A .*

REMARK 4.8. Proposition 4.6 allows one to reduce to the case of a complete noetherian local ring. By a structure theorem of Cohen³, such a ring is a quotient of a regular ring. Proposition 4.3 then allows one to assume A regular. As we shall see further on, this remark

²N.D.E. one must understand here the tensor product $\hat{I} = I \otimes_A \hat{A}$ (cf. lemma 4.5), namely I equipped with its canonical $\hat{A}$ -module structure, and not the $\mathfrak{m}$ -adic completion. For example, if p is a prime number and $A = \hat{A} = \mathbf{Z}_p$ is the ring of p -adic integers. Then, the injective envelope of the residue field $k = \mathbf{Z}/p\mathbf{Z}$ is the discrete $\mathbf{Z}_p$ -module $\mathbf{Q}_p/\mathbf{Z}_p$, whose completion for the p -adic topology is zero.

³N.D.E. see Cohen I.S., "On the structure and ideal theory of complete local rings", *Trans. Amer. Math. Soc.* **59** (1946), p. 54–106.

allows an explicit computation of the dualizing module⁴; we shall nevertheless prove theorem 4.7 by other means.

Recollections. Before proving the theorem, we make a few recollections on the notion of *injective envelope*. Cf. Gabriel, Thesis, Paris 1961, *Des Catégories Abéliennes*, ch. II §5.

Let $\mathcal{C}$ be an abelian category in which the $\varinjlim$ exist and are exact⁵ (e.g. $\mathcal{C}$ =category of modules). Every object M embeds in an injective object and one calls injective envelope of M any injective object containing M , minimal. One has the following properties:

- (i) Every object M has an injective envelope I .
- (ii) If I and J are two injective envelopes of M , there exists between I and J an isomorphism (in general not unique) which induces the identity on M .
- (iii) I is an essential extension of M , that is to say that $P \subset I$ and $P \cap M = \{0\}$ implies ($P = \{0\}$). Moreover if I is injective and an essential extension of M , I is an injective envelope of M .

These results being granted, to prove theorem 4.7, it evidently suffices to prove c).

PROOF. Let I be a dualizing module for A . Then I is injective and $\text{Hom}_A(k, I)$ is isomorphic to k . Composing the isomorphism $k \simeq \text{Hom}_A(k, I)$ with the inclusion

$$\text{Hom}_A(k, I) \hookrightarrow \text{Hom}_A(A, I) \simeq I,$$

one obtains the inclusion

$$k \hookrightarrow I.$$

Let us show that I is an injective envelope of k . Let J be an injective module such that

$$k \subset J \subset I.$$

As J is injective, there exists an injective A -submodule J' of I such that $I = J \oplus J'$. Let us show that $\text{Hom}_A(k, J') = 0$. One has

$$\text{Hom}_A(k, I) \simeq \text{Hom}_A(k, J) \oplus \text{Hom}_A(k, J');$$

$\text{Hom}_A(k, J)$ is a vector subspace of $\text{Hom}_A(k, I) \simeq k$ not reduced to zero (since it contains the inclusion $k \subset J$), hence $\text{Hom}_A(k, J) \simeq k$ and consequently $\text{Hom}_A(k, J') = 0$.

Arguing by induction on the length one deduces that $\text{Hom}_A(M, J') = 0$ for every A -module M of finite length; as I is the inductive limit of the modules $\text{Hom}(A/\mathfrak{m}^n, I)$ (cf. proposition 1.3) which are of finite length by hypothesis, the projection $I \rightarrow J'$ is zero, and consequently $J' = 0$.

Conversely, let I be an injective envelope of k . To see that I is a dualizing module, it suffices to show by 2.1 and 3.1 (ii) that $V = \text{Hom}_A(k, I)$ is isomorphic to k . Now one has the double inclusion

$$k \subset V \subset I;$$

V is a vector space over k which decomposes as the direct sum of k and of a vector subspace V' of I such that $V' \cap k = 0$. Now I is an essential extension of k , whence $V' = 0$ and $V = k$. $\square$

COROLLARY 4.9. *Let A be a noetherian local ring; every dualizing module for A is locally artinian.*

⁴This was the method followed by Grothendieck (in 1957). The method via injective envelopes which follows is due, it seems, to K. Morita, "Duality for modules and its applications to the theory of rings with minimum conditions", *Sc. Rep. Tokyo Kyoiku Daigaku* **6** (1958/59), p. 83-142. Morita's work is moreover independent of that of Grothendieck and well prior to the present seminar, and is not limited to the case of commutative base rings.

⁵**N.D.E.** of course, it is the small filtered inductive limits that are assumed to be exact; one should also assume the existence of a generator. Cf. [?]. As for the category of modules, which is sufficient for our purposes, one may also refer to Chapter 10 of Bourbaki's *Algebra*.

PROOF. Let I be a dualizing module; it is an injective envelope of k . Let us use the notations and the result of corollary 2.2. One has

$$k \subset H_{\mathfrak{m}}^0(I) \subset I,$$

and $H_{\mathfrak{m}}^0(I)$ is injective. One deduces that $I = H_{\mathfrak{m}}^0(I)$ and hence that I is locally artinian.⁶ $\square$

5. Consequences of the theory of dualizing modules

The functor

$$T = \mathrm{Hom}_A(\ , I) : \underline{\mathcal{C}}_{\mathfrak{m}} \longrightarrow \underline{\mathcal{C}}_{\mathfrak{m}}$$

is an anti-equivalence. Indeed $T \circ T$ is isomorphic to the identity functor and the argument is formal from there.

One deduces from this the usual properties of the notion of orthogonality:

Let $M^* = \mathrm{Hom}_A(M, I) = T(M)$ and let $N \subset M$ be a submodule. One calls *orthogonal* of N the submodule N' of M^* formed of the elements of M^* vanishing on N . One thus obtains a bijection between the set of submodules of M and the set of submodules of M^* , which reverses the notion of order.

One has in particular:

- $\mathrm{long}_M N = \mathrm{colong}_{M^*} N'$.
- To monogenic modules, i.e. such that $M/\mathfrak{m}M$ is of dimension 0 or 1, there correspond in the duality modules whose socle is of length 0 or 1.
- If A is artinian, the ideals of A correspond to the submodules of I .

etc.

Let A be a noetherian local ring, DA the category of A -modules M such that, for every $n \in \mathbf{N}$, $M^{(n)} = M/\mathfrak{m}^{n+1}M$ is of finite length and such that $M = \varprojlim_n M^{(n)}$, and let $\hat{A}$ be the completion of A . The restriction of scalars functor and the completion functor are *equivalences* quasi-inverse to one another between DA and $D\hat{A}$, which commute up to isomorphism with the formation of the abelian groups underlying the modules under consideration. Let us denote by CA the category of locally artinian A -modules with socle of finite dimension.

PROPOSITION 5.1. *Let A be a noetherian local ring and let I be a dualizing module for A . The functors*

$$\mathrm{Hom}_A(\ , I) : (CA)^{\circ} \longrightarrow DA$$

and

$$\mathrm{Hom}_{\hat{A}}(\ , I) : DA \longrightarrow (CA)^{\circ}$$

are equivalences of categories, quasi-inverse to one another.

Moreover, if one transports these functors by the equivalences of categories between DA and $D\hat{A}$ on the one hand, and CA and $C\hat{A}$ on the other hand, one finds the functor $\mathrm{Hom}_{\hat{A}}(\ , I)$.

PROOF. Let $X \in \mathrm{Ob} CA$. By definition, one has:

$$X = \varinjlim_{k \in \mathbf{N}} X_k, \quad X_k = \mathrm{Hom}_A(A/\mathfrak{m}^{k+1}, X),$$

hence

$$\mathrm{Hom}_A(X, I) = \varprojlim \mathrm{Hom}_A(X_k, I).$$

Hence $Y = \varprojlim X_k$ is an $\hat{A}$ -module of finite type as follows from EGA 0_I 7.2.9. One remarks in this connection that $D\hat{A}$ is also the category of $\hat{A}$ -modules of finite type or, if one prefers, that DA is the category of complete A -modules of finite type over $\hat{A}$. Let then Y be such a module, let $f : Y \rightarrow I$ be an $\hat{A}$ -homomorphism. The image of f is a submodule of finite type,

⁶N.D.E. as one has already seen, one can also simply observe that I is the inductive limit of the modules $\mathrm{Hom}_A(A/\mathfrak{m}^n, I)$.

hence is annihilated by $\mathfrak{m}^k$ for some k ; indeed every $x \in I$ is annihilated by a power of $\mathfrak{m}$. Hence f factors through $Y/\mathfrak{m}^k Y$, whence it follows that

$$\begin{aligned} \mathrm{Hom}_{\hat{A}}(Y, I) &= \varinjlim_k \mathrm{Hom}_{\hat{A}}(Y^{(k)}, I) \quad \text{with } Y^{(k)} = Y/\mathfrak{m}^{k+1}Y \\ &= \varinjlim_k (Y^{(k)})^* \end{aligned}$$

belongs to $\mathrm{Ob} \, CA$. Whence it follows immediately that the two functors of the statement are quasi-inverse to one another. $\square$

It follows from the preceding considerations that one changes nothing in the categories or the functors under consideration, nor in the abelian groups underlying the modules under consideration, by replacing A by $\hat{A}$; proposition 5.1 may then be stated as follows:

The restriction of the functor $\mathrm{Hom}_{\hat{A}}(_, I)$ to the category of $\hat{A}$ -modules of finite type takes its values in the category of locally artinian $\hat{A}$ -modules with socle of finite dimension, and admits a quasi-inverse functor, which is the restriction of the functor $\mathrm{Hom}_{\hat{A}}(_, I)$. On the intersection of these two categories, these two functors coincide (obviously!) and establish an auto-duality of the category of $\hat{A}$ -modules of finite length.

EXAMPLE 5.2 (Macaulay). Let A be a local ring with residue field k . Let k_0 be a subfield of A such that k is finite over k_0 , $[k : k_0] = d$. Every A -module of finite length can be regarded as a k_0 -vector space of finite dimension equal to $d \cdot \mathrm{long}(M)$. The functor T :

$$M \longrightarrow \mathrm{Hom}_{k_0}(M, k_0)$$

is then exact and preserves length, hence is dualizing for A . The associated dualizing module is therefore:

$$A' = \varinjlim_n \mathrm{Hom}_{k_0}(A/\mathfrak{m}^n, k_0),$$

it is the topological dual of A equipped with the $\mathfrak{m}$ -adic topology.

EXAMPLE 5.3. Let A be a noetherian local ring *regular of dimension n* . Let $\mathfrak{m}$ be its maximal ideal, let k be its residue field. There exists a regular system of parameters, $(x_1, x_2, \dots, x_n)$, which generates $\mathfrak{m}$, and which is an *A -regular sequence*. One can therefore compute the $\mathrm{Ext}_A^i(k, A)$ by the Koszul complex; one finds:

$$\begin{aligned} \mathrm{Ext}_A^i(k, A) &= 0 \quad \text{if } i \neq n, \\ \mathrm{Ext}_A^n(k, A) &\simeq k. \end{aligned}$$

The depth of A being n , for every M annihilated by a power of $\mathfrak{m}$, $\mathrm{Ext}_A^i(M, A) = 0$ if $i < n$; moreover $\mathrm{Ext}_A^i(M, A) = 0$ if $i > n$ because the global cohomological dimension of A is equal to n . Hence $\mathrm{Ext}_A^n(_, A)$ is exact and moreover $\mathrm{Ext}_A^n(k, A) \simeq k$; it follows that:

PROPOSITION 5.4. *If A is a regular noetherian local ring of dimension n , the functor*

$$M \longrightarrow \mathrm{Ext}_A^n(M, A)$$

is dualizing. The associated dualizing module is

$$I = \varinjlim_r \mathrm{Ext}_A^n(A/\mathfrak{m}^r, A),$$

it is isomorphic to $H_{\mathfrak{m}}^n(A)$ (Exposé ??, Th. ??)⁷.

⁷Let A be a ring, J an ideal of A , M an A -module, $i \in \mathbf{Z}$; one then sets $H_J^i(M) = H_Y^i(X, F)$, where $X = \mathrm{Spec}(A)$, $Y = V(J)$ and $F = \tilde{M}$.

REMARK 5.5. If A satisfies both the hypotheses of the two preceding examples, the two dualizing modules found are isomorphic. Suppose for example that A is regular of dimension n , complete and of equal characteristics. There then exists a field of representatives, say K . If one chooses a system of parameters $(x_1, \dots, x_n)$ of A , one can construct an isomorphism between A and the ring of formal power series: $K[[T_1, \dots, T_n]]$; whence, as we shall see, an *explicit* isomorphism between the two dualizing modules

$$v: H_{\mathfrak{m}}^n(A) \longrightarrow A'.$$

One can find an *intrinsic* interpretation of this isomorphism with the help of the module $\Omega^n = \Omega^n(A/K)$ of completed relative differentials of maximal degree. Indeed, one knows that Ω^n admits a basis formed of the element $dx_1 \wedge dx_2 \cdots \wedge dx_n$.

Whence an isomorphism

$$u: H_{\mathfrak{m}}^n(\Omega^n) \longrightarrow H_{\mathfrak{m}}^n(A).$$

A remarkable fact is then that the composite

$$vu = w: H_{\mathfrak{m}}^n(\Omega^n) \longrightarrow A'$$

no longer depends on the choice of the system of parameters and commutes with change of the base field.

To construct v one computes $H_{\mathfrak{m}}^n(A)$ by means of the Koszul complex associated to the x_i , one finds:

$$H_{\mathfrak{m}}^n(A) = \varinjlim_r A/(x_1^r, \dots, x_n^r);$$

where the transition morphisms are defined as follows: set $I_r = A/(x_1^r, \dots, x_n^r)$; let $e_{a_1, \dots, a_n}^r$ be the image of $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ in I_r . The $e_{a_1, \dots, a_n}^r$, for $0 \leq a_i < r$ form a basis of I_r .

This being said, if s is an integer, the transition morphism

$$t_{r, r+s}: I_r \longrightarrow I_{r+s}$$

is multiplication by $x_1^s x_2^s \cdots x_n^s$, hence:

$$u_{r, r+s}(e_{a_1, \dots, a_n}^r) = e_{a_1+s, \dots, a_n+s}^{r+s}.$$

Note that the datum of an A -homomorphism w of an A -module M into A' is equivalent to the datum of a K -linear form $w': M \rightarrow K$ which is *continuous* on the submodules of finite type. In the case $M = H_{\mathfrak{m}}^n(\Omega^n)$, the definition of w is therefore equivalent to that of a linear form

$$\rho: H_{\mathfrak{m}}^n(\Omega^n) \longrightarrow K,$$

called *residue form*⁸. To construct ρ , it suffices to define forms $\rho_r: I_r \rightarrow K$ which glue together, and one will take

$$\rho_r(e_{a_1, \dots, a_n}^r) = \begin{cases} 1 & \text{if } a_i = r-1 \text{ for } 1 \leq i \leq n \\ 0 & \text{otherwise.} \end{cases}$$

⁸For a more detailed study of the notion of residue, Cf. R. Hartshorne, *Residues and Duality*, Lect. Notes in Math., vol. 20, Springer, 1966.